Averil Prost | PhD student in Applied Mathematics

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Formation

PhD in Applied Mathematics *since Oct. 2022*
LMI - Laboratory of Mathematics of INSA
- Control problems over measure spaces, with N. Forcadel and H. Zidani

Engineering diploma in Applied Mathematics *2017-2022*
National Institute of Applied Sciences - INSA Rouen
- Functional and numerical analysis for PDEs, deterministic control theory

Master in Fundamental and Applied Mathematics *2021–2022*
University of Rouen Normandie
- Viscosity solutions, Markov processes, particle systems

Integration of Graduate School MINMACS *2021–2022*
Excellence scolarship in M2

Internships

CEMRACS - Vlasov-Poisson plasma sheath (M. Badsi, M. Mehrenberger) **CIRM**
Comparison of semi-Lagrangian, finite differences and fixed-point methods for a bispecies plasma sheath with absorbing boundary conditions. *15th July - 31th August 2022*

Numerical methods for Hamilton-Jacobi equations (O. Bokanowski) **Lab. J.L. Lions**
Semi-Lagrangian scheme for obstacle problems with neural networks approach. *1st March - 15th July 2022*
https://arxiv.org/abs/2210.04300

Implicit-explicit scheme for the wave equation (A. Impériale) **CEA Saclay**
Multi-scale semi-implicit scheme in inhomogeneous media, with finite elements. *June - August 2021*
https://www.github.com/averil-prost/Wonderbubbleland

Projects

Introduction to nonlinear control theory *Master's thesis*
- Optimality conditions and adjoint operators for Nemytskii equations

https://github.com/averil-prost/NonLinearite

Introduction to numerical simulation of Navier-Stokes *Final engineering project*
- Finite element schemes for Stokes and incompressible Navier-Stokes

Introduction to geometric PdEs on graphs *4th year project*
- Definition and simulation of Laplace and Eikonal equations

Aptitudes

- C/C++/Julia
- Python/Matlab
- LaTeX/LyX
- Linux/Windows
- Paraview
- OpenMP/MPI
- French: native speaker
- English: C1, 990/990 at TOEIC (2021)
- Spanish: A1